

Supplemental Material for “Scaling Advantage in Approximate Optimization with Quantum Annealing”

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I. EXPERIMENTS

We performed QA experiments on the D-Wave Advantage 4.1 QPU accessed through the D-Wave Leap cloud interface, featuring the Pegasus hardware graph [1].

All previous QAC results were obtained using the “Chimera” hardware graph of the previous generation of D-Wave QPUs, which has degree 3 and no odd-length loops. The features of the induced Pegasus logical graph permit the benchmarking of significantly harder spin-glass instances than was possible on Chimera.

We used both the standard, “slow” annealing schedule and the newer, “fast” schedule. These schedules are depicted in Fig. 1. We note that despite the name “fast schedule,” these schedules can also be operated as “slow” anneals, like the standard schedules; the latter, however, cannot be operated fast. The technical features of the annealing schedules are described in the D-Wave documentation: https://docs.dwavesys.com/docs/latest/c_qpu_annealing.html.

The fast schedules are determined by the ramping of qubit fluxes, while the standard schedules arise from a linear growth of the persistent current, resulting in the different shapes of the fast annealing schedules shown in Fig. 1, and the need for longer anneal times at larger problem sizes to reduce diabatic transitions. We believe that this explains why the t_f values in Fig. 1 of the main text are larger for the fast sweeps than for the slow sweeps for sufficiently large system sizes.

II. QAC ENCODING AND INDUCED LOGICAL GRAPH

Fig. 2 illustrates the encoding and the induced logical graph used in our experiments.

III. QAC PENALTY STRENGTH OPTIMIZATION

We restricted the penalty coupling strengths to the set $J_p \in \{0.1, 0.2, 0.3, 0.4\}$ in our experiments to reduce resource requirements for parameter optimization. In most

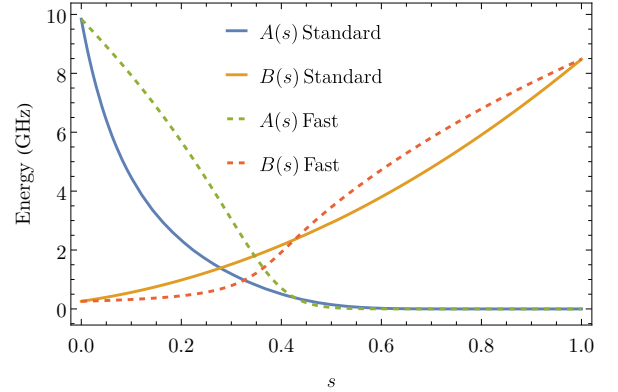


FIG. 1. Standard and fast annealing schedules available via Leap for the D-Wave Advantage QPU. The annealing parameter $s = t/t_f \in [0, 1]$, where t_f is chosen by the user.

cases where we find an optimal annealing time, the corresponding optimal penalty coupling is typically $J_p \approx 0.2$ to 0.3 , with 0.3 being frequently optimal for the fast annealing schedule at $\varepsilon \geq 1.25\%$.

In the context of the reduction of the KZ exponent μ shown in Fig. 3 of the main text, we found that this reduction is lost when $J_p = 0.2$ (not shown), suggesting that $J_p = 0.1$ is optimal in the sense of diabatic error suppression.

IV. GROUND STATE ENERGIES

We solved all S28 instances to optimality using Gurobi 10 within feasible runtime, except for 7 instances of size $L = 15$. For these remaining instances, Gurobi proved an optimality gap of at most 1.2%, meaning the lowest energy found is guaranteed to be no greater than 1.2% above the true ground state. Furthermore, the lowest energies found by PT-ICM were no higher than those found by Gurobi for all instances. Thus, we assigned the ground state energies of the remaining instances to the values found by PT-ICM with high confidence. As we use a median over instances as our summary statistic, our conclusions are unaffected by the possibility that the ground state energy was never reached for such few

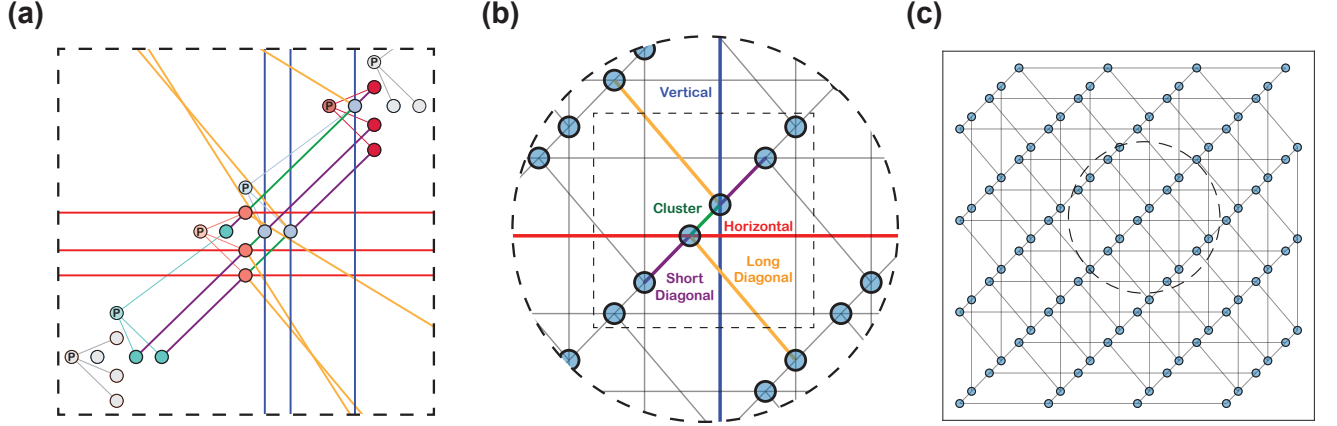


FIG. 2. Embedding of QAC on the Pegasus hardware graph. (a) Close-up of the embedding of three clusters of two coupled logical qubits each. Physical qubits are denoted by circles and couplings by lines. Penalty qubits are marked by a P and the three penalty couplings (thin lines) to their corresponding physical data qubits. Thick lines indicate the physical couplings between the corresponding physical qubits of different logical qubits. Only a subset of all possible couplings are shown. (b) A zoomed-out view from (a) showing the logical qubits (circles) and the logical couplings induced by the QAC embedding. Thick lines indicate the logical couplings shown in (a), whose type is colored by direction: horizontal/vertical/diagonal (long or short). (c) A zoomed-out view from (b) showing a 4×4 induced logical graph of QAC. The logical graph is equivalent to a honeycomb graph with additional non-planar bonds. The induced logical graph of the D-Wave Advantage 4.1 QPU has a maximum of 1322 logical qubits; the largest available side length is $L = 15$.

instances.

V. PT-ICM PARAMETER OPTIMIZATION

We first ran every replica once for $N_{\text{sw}}^{\text{max}} = 500,000$ sweeps to ensure the ground state energy was reached. For the largest size $L = 15$, we determined $N_{\text{sw}}^{(90\%)}$, the number of sweeps that were required for 90% of the instances to reach their lowest recorded energy, where we found $N_{\text{sw}}^{(90\%)} \approx 31,000$ for the S28 instances. As this was significantly less than $N_{\text{sw}}^{\text{max}}$, we considered the ground states for these instances as “validated” by PT-ICM to calculate median quantities over the instances. We finally ran PT-ICM 100 times for each instance, setting $N_{\text{sw}} = N_{\text{sw}}^{(90\%)}$. This yields an empirical cumulative density function for $p_{E \leq E_0 + \varepsilon}$ as a function of the runtime of PT-ICM. The TTE is then evaluated for each instance by optimizing over the runtime of the PT-ICM repetition (where t_f is now the time needed to reach the target rather than the annealing time).

The scaling of PT-ICM is ideally evaluated using the parameters that best optimize the TTE for each disorder realization, instance size, and target ε . However, a rigorous optimization of the number and choice of replica temperatures for all target ε 's and system sizes is computationally infeasible. To ensure our results hold for any choice of reasonably optimized temperatures, we repeated the TTE evaluation with four temperature sets summarized in Table I, which includes both logarithmically-spaced and feedback-optimized [2] tem-

		S28		
Set	N_T	$\beta_{\min}$	$\beta_{\max}$	N_{icm}
1	32	0.1	5.0	8
2	24	0.2	10.0	6
3	32	0.2	10.0	8
4	32	0.1	20.0	8

TABLE I. Temperature sets used for PT-ICM for S28 instances, where $\beta_{\min}$ is the hottest inverse temperature, $\beta_{\max}$ is the coldest inverse temperature, and N_{icm} is the number of low temperature (largest β) subject to ICM moves. The temperatures are logarithmically spaced in sets 1 and 4. The temperatures in sets 2 and 3 were feedback-optimized with initially logarithmically spaced temperatures.

peratures. The TTE for a given disorder, instance size, and energy target was chosen from the best TTE out of the four temperature sets, illustrated in Fig. 3. At the optimality gap targets of interest, most temperature sets' differences are not appreciable and are unlikely to affect our conclusions. A more comprehensive but computationally expensive optimization of PT-ICM would involve a grid search over a range of the parameters $\beta_{\min}$, $\beta_{\max}$, and N_{icm} .

VI. SOFTWARE IMPLEMENTATION

Our PT-ICM implementation is available as the TAMC software package [3]. Our implementation of the QAC and C3 embeddings for the D-Wave Advantage graph is

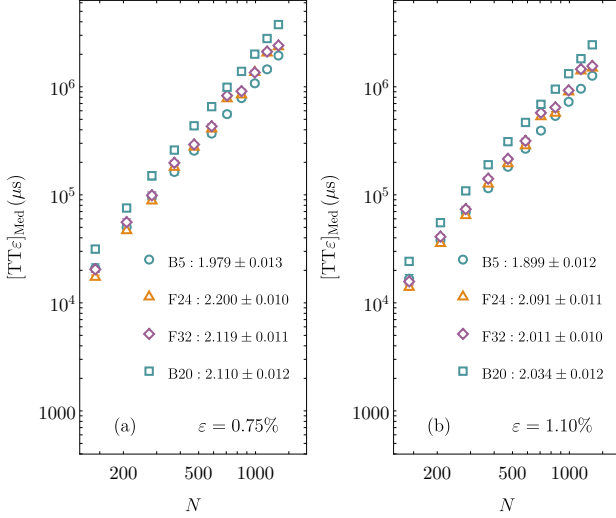


FIG. 3. Time-to-epsilon of the different temperature sets used for PT-ICM and their individual scaling for $\varepsilon = 0.75\%$ (left) and $\varepsilon = 1.1\%$ (right). The four sets correspond, in order, to the sets shown in Table I. B5: Logarithmically-spaced with $\beta_{\max} = 5$. F24: Feedback-optimized with 24 temperatures. F32: Feedback-optimized with 32 temperatures. B20: Logarithmically-spaced with $\beta_{\max} = 20$.

available as part of the PegasusTools Python package [4]. All quantum annealing methods were prepared, submitted to D-Wave API, and analyzed with this package.

We also implemented and considered the performance of simulated annealing but do not present its results here as this algorithm was not competitive at large problem sizes. Our PT-ICM implementation is a general-purpose solver written in the Rust programming language, targeting CPU execution, and accepts instances with any connectivity. This is in contrast to previous studies that have used simulated annealing or the Hamze-de Freitas-Selby algorithm solvers that are specialized for problems defined on the Chimera graph [5, 6].

VII. STATISTICAL METHODS

For a given problem size and target optimality gap, we calculate the median $\text{TT}\varepsilon$ from the quantum annealing data using a three-step Bayesian bootstrap procedure over the levels of readout samples, gauges (spin-reversal transformations at the hardware level), and instances: (1) the success probability for each gauge is re-sampled from a beta distribution for N_{samp} samples per gauge, (2) the statistical weight of each gauge is sampled from a Dirichlet distribution of length N_{gauge} to take the weighted average success probability for each instance, and (3) the statistical weight of each instance is sampled from a Dirichlet distribution of length N_{inst} to take the weighted median. We performed our quantum annealing experiments with $N_{\text{samp}} = 1000(\text{QAC})/3000(\text{C3})$,

$N_{\text{gauge}} = 10$, and $N_{\text{inst}} = 125$. We performed $N_{\text{boots}} = 200$ bootstrap samples per size and energy density target pair and found the annealing time and penalty strength that resulted in an optimal $\text{TT}\varepsilon$ for each bootstrap sample. The distribution of the optimal median $\text{TT}\varepsilon$ values and the distribution of optimal annealing parameters are the two final products of this sampling procedure shown in Fig. 2 of the main text, with 1σ error intervals for the optimal $\text{TT}\varepsilon$.

Next, we formulate a null hypothesis and compute P values as follows. The null hypothesis is that the optimal annealing time t_f is the minimum accessible $t_f^{\min} = 0.5 \mu\text{s}$ for the standard schedule or $t_f^{\min} = 5 \text{ ns}$ for the fast schedule, i.e., that the true optimal annealing time is not above $t_f^{\min}$. The P value is the empirical number of bootstrap samples whose optimal t_f was $0.5 \mu\text{s}$ (or 5 ns for the fast schedule), out of 200 samples. Filled circles in Fig. 2 of the main text mean $P < 0.05$ for the probability that $t_f = 0.5 \mu\text{s}$ (or 5 ns for the fast schedule) in the bootstrap sample, while open circles mean $P < 0.20$. The filled circles show which points have the highest confidence that the optimal annealing time is not below $0.5 \mu\text{s}$ (or 5 ns for the fast schedule).

To compare the quality of fitting with either an exponential or polynomial fit, we begin by fitting the $\text{TT}\varepsilon$ to a linear model containing both N and $\log_{10} N$, i.e.,

$$\log_{10} \text{TT}\varepsilon(N) = a + \alpha \log_{10} N + \beta N, \quad (1)$$

which corresponds to a general poly-log model $\text{TT}\varepsilon = cN^\alpha 10^{\beta N}$ where $a = \log_{10} c$. We then perform F -statistic tests on the sum of squared errors (SSE) for the poly-log model ($\text{SSE}_{\alpha,\beta}$) and for its two possible nested models: the polynomial scaling fit (SSE_α , from fitting α constraining $\beta = 0$) and the exponential scaling fit (SSE_β , from fitting β constraining $\alpha = 0$). From these SSE values, we evaluate the corresponding F -statistics

$$F_i = \frac{\text{SSE}_i - \text{SSE}_{\alpha,\beta}}{\text{MSE}_{\alpha,\beta}}, \quad i = \alpha, \beta \quad (2)$$

for the polynomial and exponential models. The p -values of the F -statistics are obtained from a Type I sum-of-squares ANOVA analysis. The null hypothesis $H_0(i)$ of the F -test is that the poly-log model is not a stronger fit than model i [quantified by the reduction in SSE by allowing one extra parameter to be fit, relative to the mean squared error (MSE)], for each $i = \alpha, \beta$. When the F_i statistic is significant, this null hypothesis is rejected, which implies model i is less suitable than the full poly-log model. We interpret the rejection of the null hypothesis $H_0(\beta)$ for the exponential fit along with the failure to reject the null hypothesis $H_0(\alpha)$ for the polynomial fit as evidence that the polynomial fit is a stronger explanatory model for the scaling than the exponential fit. Conversely, we also interpret the rejection of $H_0(\alpha)$ with the failure to reject $H_0(\beta)$ to be evidence that the exponential fit is a stronger explanatory model than the polynomial fit.

We performed the above test on the fits for PT-ICM, C3, and QAC-f and overwhelmingly found that a purely exponential fit is rejected in favor of a poly-exponential fit, with p -values for F_β no more than 1.73×10^{-8} (PT-ICM), 1.0×10^{-3} (C3), and 1.6×10^{-4} (QAC-f). In contrast, the p -values for F_α were no less than 0.01 across all values of ε and clearly not rejectable for $\varepsilon = 1.1\%$ and 1.25% . As remarked in the main text, while this does not rule out that a poly-exponential scaling is possible, the failure to reject $H_0(\alpha)$ at the current range of system sizes is more consistent with a exponential scaling for all our methods, with the exception of C3-f at smaller system sizes (see Fig. 2 in the main text).

VIII. NON-UNIVERSALITY OF THE OBSERVED KZ CRITICAL EXPONENT

The Binder cumulant U is well-known to provide a statistical signature of phase transitions. Under the dynamic finite-size scaling ansatz, $U(L, t_f)$ is expected to collapse onto a common curve for all system sizes N when t_f is rescaled by $L^{-z-1/\nu}$, where ν and z are the correlation length and dynamic critical exponents, respectively. This reflects the KZ ansatz: the annealing time required for the system to remain adiabatic up to a correlation length of L scales as

$$t_f(L) \sim L^\mu, \quad \mu = z + \frac{1}{\nu}. \quad (3)$$

While the collapse seen in Fig. 4 of the main text is visually convincing, the error estimates for U are, unfortunately, too large to determine the KZ exponent or to extract ν and z . In addition, we do not observe that the extracted KZ exponent is universal: the estimate for μ that best collapses the Binder cumulant for binomial (as opposed to S28) spin glass instances is $\mu \approx 9 \pm 1$ for C3, and is reduced to $\mu \approx 7.5 \pm 0.6$ with QAC at $\lambda = 0.1$. Thus, while the scaling ansatz yields a useful complementary way to quantify the advantage of QAC in terms of sampled quantities in addition to the $\text{TT}\varepsilon$ metric, the lack of universality and the possibility that the spin-glass transition temperature is zero [7] do not clearly support a universal critical description of the annealing dynamics, quite unlike the conclusions of Refs. [8, 9].

IX. TIME-TO-RESIDUAL ENERGY

An alternative time metric is the residual energy density from the ground state; in the main text, we focus on the optimality gap due to its relevance to benchmarking approximate optimization algorithms.

Ref. [9] performed energy decay measurements of QA dynamics in 3D spin glass instances as captured through the residual energy, a dimensionless quantity,

$$\rho = \frac{\langle H_z \rangle - E_0}{JN}, \quad (4)$$

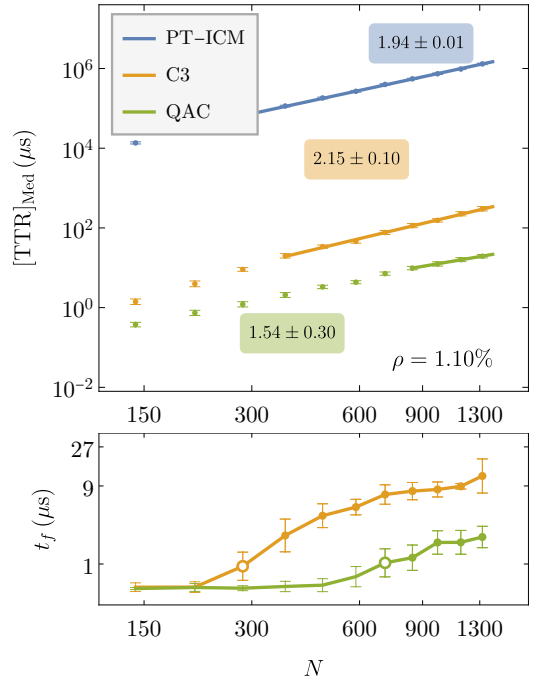


FIG. 4. Time-to- ρ performance and scaling of S28 instances, which are comparable to those using the time-to-epsilon metric.

where N is the number of spins and J is the characteristic energy scale of the Ising Hamiltonian (which is simply 1 for all of our instance classes). This motivates an alternative measure for approximate optimization, which we call *time-to-residual energy*, or *time-to- ρ* ,

$$\text{TTR}(N) = t_f \frac{\log(1 - 0.99)}{\log(1 - p_{E \leq E_0 + \rho JN})}, \quad (5)$$

where now ρ sets the target energy difference from the ground state in units of JN , rather than E_0 as in the case of $\text{TT}\varepsilon$. Nevertheless, both ε and ρ are targets that grow in proportion to the problem size or number of variables (in finite dimensions). The TTR analog of Fig. 2 of the main text at $\rho = 1.1\%$ is shown in Fig. 4. While the precise scaling exponents α of C3 and QAC vary slightly, the optimal annealing times are nearly identical. The trends of the scaling exponents as functions of either ρ or ε are also qualitatively similar. In finite-dimensional spin glasses, the variance of E_0 across instances appears to scale with $\sqrt{N}$ in the thermodynamic limit [10]. Hence, E_0/NJ converges to a constant, and there is effectively no distinction between ε and ρ in the thermodynamic limit.

X. CLASSICAL COMPLEXITY OF APPROXIMATE OPTIMIZATION

Under the TTR metric, it can be shown that the approximate optimization of finite-range spin glasses has, in

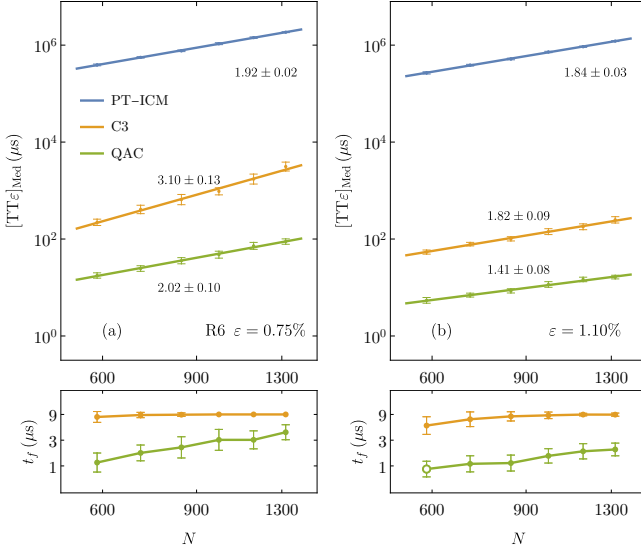


FIG. 5. Time-to-epsilon scaling of spin glass instances with R6 disorder for (a) $\varepsilon = 0.75\%$ and (b) $\varepsilon = 1.10\%$. To expedite finding the optimal TTE , the annealing times were limited to the range $t_f \in [0.5, 9] \mu\text{s}$ and QAC was limited to $J_p = 0.2$ only.

theory, a linear scaling using a simple divide-and-conquer algorithm. Namely, for a given residual energy target ρ , an algorithm exists whose TTR scales linearly in the system size for a sufficiently large size, with a large prefactor depending on the residual energy. For 2D spin glasses on a square lattice, it can be summarized as follows:

1. Partition the instance of size $N = L^2$ into $K \times K$ subgraphs $G_{x,y}$, with $x, y \in \{1, \dots, K\}$, each with *constant* side length of $L_0 = L/K$ spins;
2. Find the local ground states for each subgraph using an exact or heuristic solver for each $L_0 \times L_0$ instance. This step has cost $C(L_0)$, potentially exponential in L_0^2 on a non-planar lattice;
3. Patch together all subgraph ground state configurations as the approximate ground state for the global Hamiltonian and return this state's energy as the approximate ground state energy $E^*(N)$. This last step requires $O(N)$ time due to the need to sum the energies of $K^2 = O(N)$ patches.

The overall complexity is, therefore, $C(L_0)O(N)$.

This algorithm optimizes *bulk* energies throughout the volume of the spin-glass at the expense of large energy violations over regions scaling with the surface area of the spin-glass, i.e., at the patch boundaries. The boundary excitations become negligible for sufficiently large sizes compared to the bulk energies, with the latter eventually reaching the desired residual energy. More precisely, up to $4L_0$ boundary couplings may be violated per volume of L_0^2 , so the residual energy density for this algorithm is upper-bounded by $4/L_0$. Thus, we estimate that the

regime of system sizes where this algorithm applies for the ρ targets we examine, e.g., 1%, would require a reliable, efficient solver for instances with at least a *sub-problem* side length of $L_0 \approx 400$, resulting in a prefactor of $2^{400} \sim 10^{120}$, which is entirely impractical. Nevertheless, this algorithm could be a starting point for parallel and quantum-classical hybrid algorithms for massive, finite-dimensional problems. Furthermore, such an algorithm could reach a similar target ε in linear time with increasing probability as the system size increases due to the decreasing variance of $E_0/(JN)$.

XI. ALTERNATIVE SPIN-GLASS DISORDER CASES

We performed a similar study of instances with binomial disorder $J = \pm 1$, using only the standard schedule. Unsurprisingly, there was no substantial difference in scaling between C3 and QAC, and additionally, we were unable to validate the optimal annealing time for QAC at $\varepsilon = 1\%$. Thus, we considered the binomial disorder instances unsuitably easy for our purposes.

In an attempt to estimate the influence of precision and ground state degeneracy in the S28 disorder, we also studied range 6 (R6) instances for which the couplings were randomly drawn from $J \in \{\pm 1/6, \pm 2/6, \pm 3/6, \pm 4/6, \pm 5/6, \pm 1\}$ [11]. The main contrast between R6 and S28 is that the minimum non-zero local field a qubit may experience is much smaller in S28 disorder. Furthermore, in the absence of the Sidon property, the R6 disorder is more susceptible to ground-state degeneracies due to floppy qubits. However, it is less likely to occur than in a smaller range disorder case such as range 3.

The TTE scaling for the R6 case is shown in Fig. 5. For a smaller $\varepsilon = 0.75\%$, the scaling of QAC is better with R6 disorder than S28 disorder (Fig. 1 of the main text), going from 2.2 to 2.0, then equalizes as ε increases. Perhaps surprisingly, the opposite is true of C3, which scales worse on R6 disorder at smaller epsilon (2.9 to 3.1) before scaling better than S28 disorder above $\varepsilon \approx 1.1\%$ (2.0 to 1.8). That is, while S28 is overall more challenging than R6 under QAC, low energy sampling of R6 is more challenging for unprotected QA than S28. This is likely caused by the worse relative precision of small coupling strengths in unprotected QA, which would not affect the S28 disorder.

XII. ADDITIONAL RESULTS FOR S28 DISORDER

We present additional S28 disorder results for different quantiles, complementing the results presented for the median quantile in the main text. Fig. 6-Fig. 8 are analogous to Fig. 2 in the main text, including additional scaling results for the median quantile at different op-

timality gap values. Fig. 9 summarizes the polynomial scaling exponent results for

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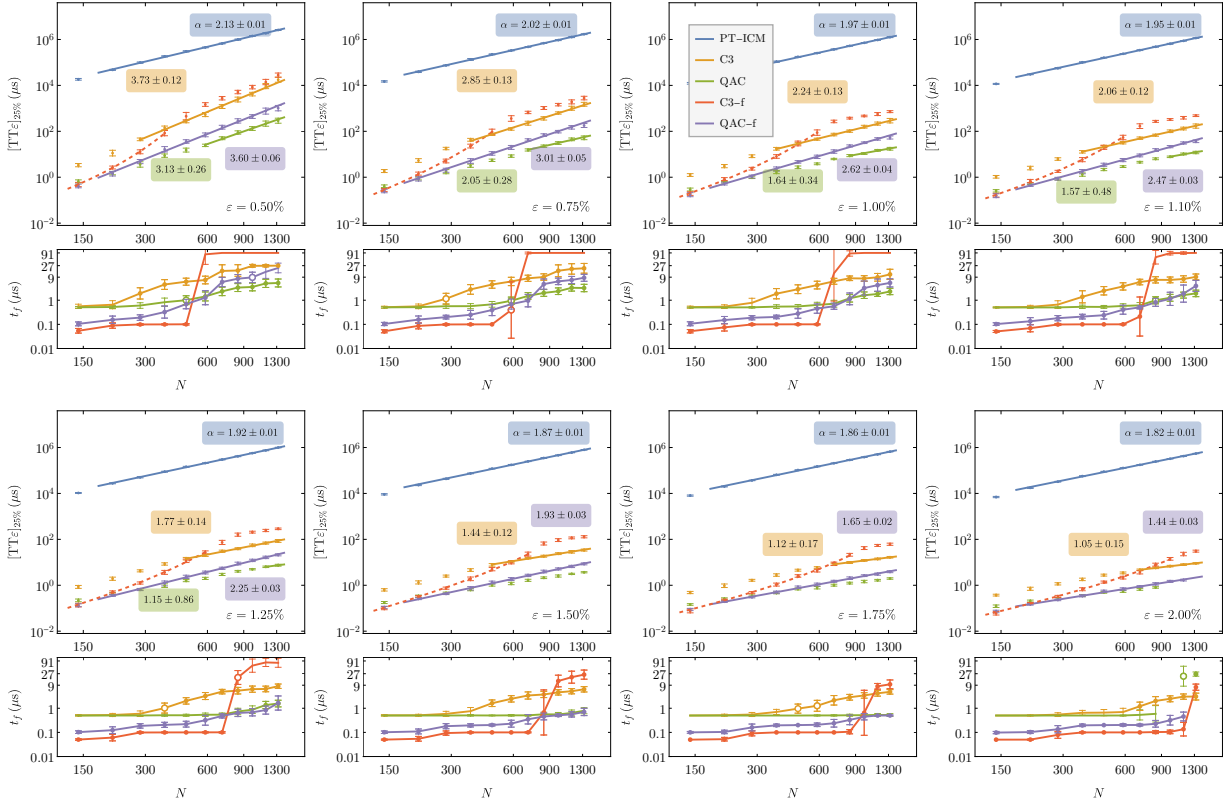


FIG. 6. Same as Fig. 2 in the main text, but for the 25th quantile ($q = 0.25\%$).

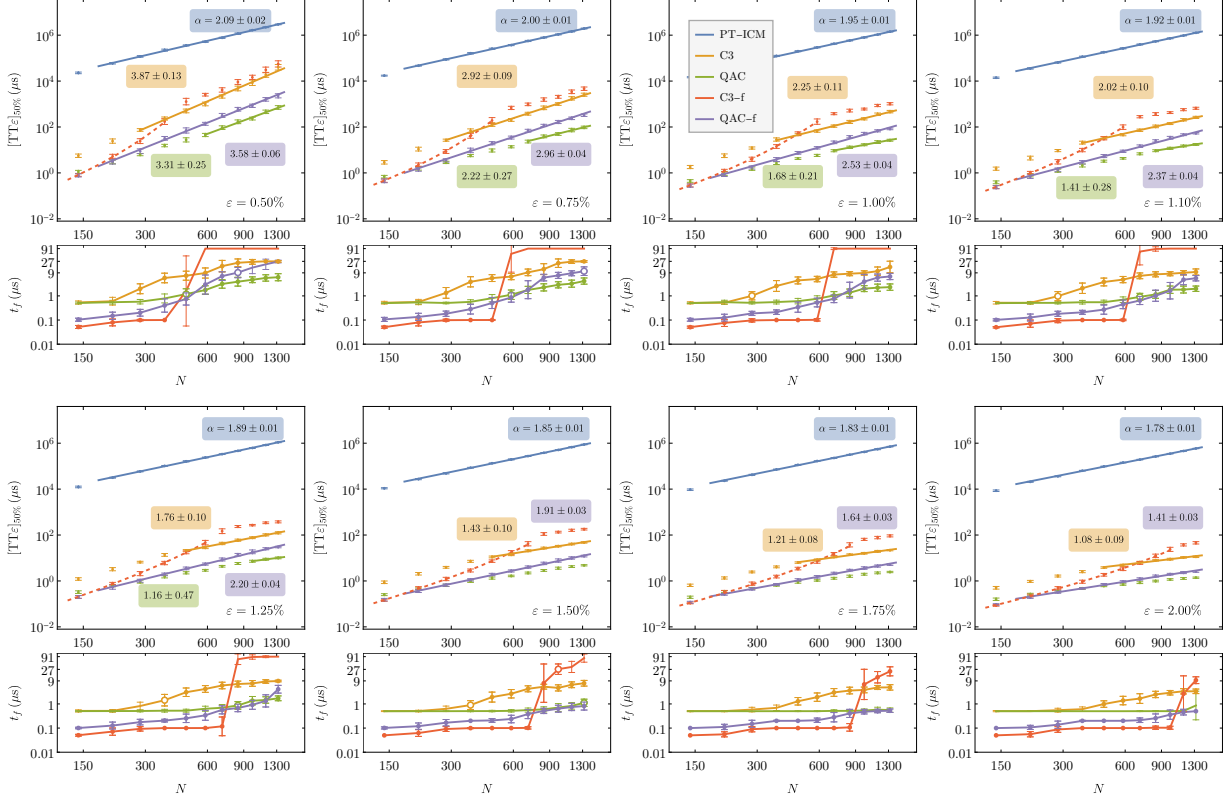


FIG. 7. Same as Fig. 2 in the main text, with additional optimality gap values for $q = 50\%$.

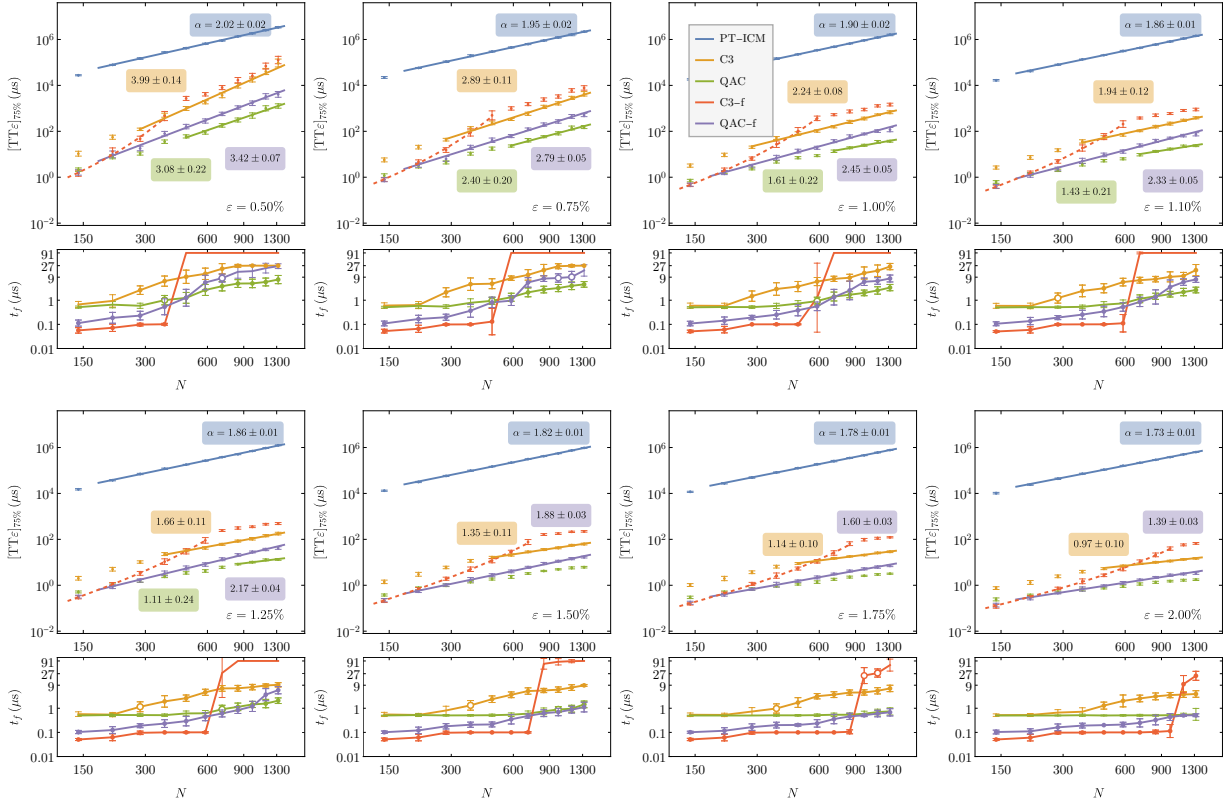


FIG. 8. Same as Fig. 2 in the main text, for $q = 75\%$.

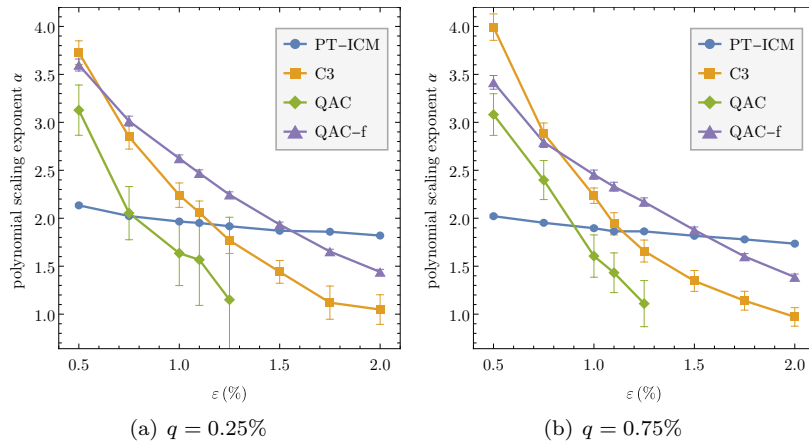


FIG. 9. Same as Fig. 3 in the main text, for the lower and upper quartiles. If the error bars are ignored then the higher the quantile, the higher the value of ε is where QAC intersects PT-ICM, marking a transition from slowdown to speedup.